

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
ML / Deep Learning	PyTorch, JAX, HuggingFace Transformers, sentence-transformers, NumPy, scikit-learn, normalizing flows, diffusion, variational inference, autodiff
LLM / Agentic / Eval	LLM agents, tool use, typed retrieval planning, deterministic + LLM extraction, cell-level provenance, RAG, RRF, cross-encoder reranking, MMR, evaluation methodology, holdout protection
Backend & Systems	FastAPI, Node.js, REST APIs, async services, Linux profiling, Apache Spark / PySpark, Spark UI, MongoDB, PostgreSQL, Redis, SQLite
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Research & Projects

AgenticSearch - Provenance-First Agentic Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, Brave Search | live demo

- Built a multi-stage **agentic web-discovery pipeline** that turns open-ended queries into structured entity tables via typed retrieval planning, tool use (search and page rerank), evidence-regime classification, deterministic extraction with LLM fallback, and output verification, returning **cell-level provenance** for every value.
- Hardened the agent with **150+ automated tests**, pipeline instrumentation, and provider routing/fallback across OpenAI and Groq to keep multi-stage extraction predictable and auditable across LLM backends.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Designed an **evaluation-improvement agent** that proposes queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

Refusal Decay - LLM Safety Mechanistic Interpretability

UMass CS 602 Research Project

Python, PyTorch, HuggingFace Transformers | graded full marks

- Ran an independent mechanistic-interpretability study of how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct; designed two causal interventions (activation patching, direction injection) with multi-seed controls and reported a clean null with bootstrap **95% CIs** and McNemar’s test, showing the signal is a readout rather than the causal lever at the tested sites.

Generative Models & Deep Learning

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented a **matrix-based reverse-mode autodiff engine from scratch** in NumPy (computation graphs, topological-sort backprop, matmul / solve / log-determinant operators), **validated against JAX** to machine precision.
- Implemented generative models from first principles in JAX, a **normalizing flow with coupling layers** and a **DDPM diffusion model**, deriving the ELBO and Gaussian-KL objectives via variational inference; benchmarked **5 architectures** across **3 optimizers** on CIFAR-10.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions** of real operational data; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.